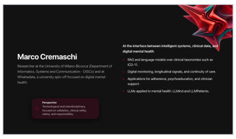
Note del relatore

Slide 2

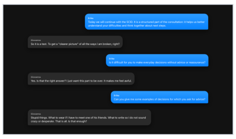


- Briefly about me.
- I am a **researcher** at the **University of Milano-Bicocca**, in the **DISCo department**, and at **Whattadata**.
- My work is at the interface between **intelligent systems**, **clinical data**, and **digital mental health**.
- I work with **RAG**, clinical taxonomies such as **ICD-11**, **digital monitoring**, and tools for **adherence**, **psychoeducation**, and **clinician support**.
- The perspective is technological and interdisciplinary: **validation**, **clinical utility**, **safety**, and **responsibility**.

Slide 3

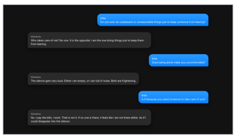
- The solutions I describe today are developed in **Whattadata**.
- Whattadata is a **university spin-off** of the **University of Milano-Bicocca**.
- It works on **data**, **models**, and **digital platforms** for **mental health**.
- The goal is not only to build **prototypes**, but to move from **design** to **field validation**.

Slide 4



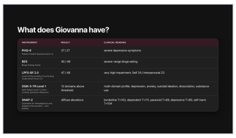
- Let us start inside a **consultation**.
- The **SCID** is a **structured clinical interview**.
- It helps the clinician ask **diagnostic questions** in a systematic way.
- Here the therapist asks **SCID-style questions**.
- Giovanna reacts with **shame** and **fear of being judged**.
- The point is not only the **answer** to the question.
- The point is the **relational reaction**.

Slide 5



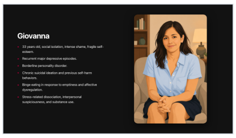
- Here the same **diagnostic area** becomes more complex.
- The patient rejects the idea that she simply needs someone to **take care of her**.
- She describes **emptiness** and **fear of disappearing**.

Slide 6



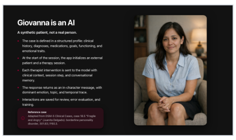
- This slide shows **questionnaires** and **clinical scales**.
- They are not the whole **diagnosis**.
- They help measure different areas: **depressive symptoms**, **binge eating**, **personality functioning**, **broad psychiatric symptoms**, and **personality traits**.
- Together, these scores define a **severe** and **complex profile**.
- They point to **high impairment** across many domains.

Slide 7



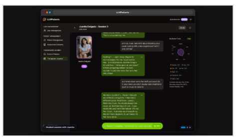
- This is the **clinical formulation** in human words.
- Giovanna is **isolated**, **ashamed**, **depressed**, and has **borderline personality disorder**.
- She also has **chronic suicidal thoughts** and a history of **self-harm**.
- This is exactly why simulation must be **supervised** and **safe**.

Slide 8



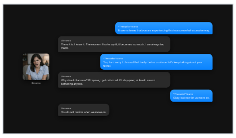
- Now the **key twist**: Giovanna is not real.
- She is a **synthetic patient**.
- This does not mean that we just asked **ChatGPT** to act as Giovanna.
- The case is defined in a **structured profile**, with **history**, **diagnoses**, **goals**, **functioning**, **medication**, and **emotional traits**.
- At the beginning of the session, the app creates the **patient** and the **therapy context**.
- Every therapist message is sent together with **clinical context**, **memory**, and the current **session step**.
- The **LLM** writes the final sentence, but the simulated patient is represented by **profile**, **state**, **memory**, and **logs**.

Slide 9



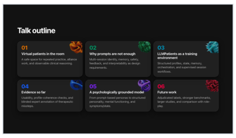
- This is **LLMPatients** in use.
- The slideshow shows the **student chat**, the **patient grid**, and the **pathway dashboard**.
- It is a tool that **simulates patients** for **training**.
- It is built to help **psychology students** develop the skills needed to conduct a **therapeutic interview**.
- Students can practise **asking questions**, **listening**, **managing ruptures**, and **repairing the relationship**.
- The conversation is **logged**, so a lecturer can later inspect **difficult turns** and use them for **supervision**.

Slide 10



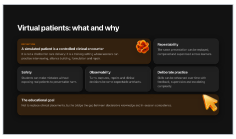
- This was my **conversation with Giovanna**.
- Here I was a **bad therapist**.
- I made several **mistakes**.
- First, I **minimized her experience** by saying it was excessive.
- Then I tried to move forward too quickly, without **repairing the rupture**.
- After these mistakes, Giovanna **stopped answering** me.
- This is a very good example of what we should **not do** in a **therapeutic interview**.

Slide 11



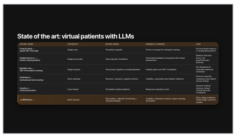
- I will move through **six points**.
- First, why **virtual patients** are useful.
- Second, why **prompt-only personas** are not enough.
- Then I show **LLMPatients**, the **early evidence**, the **grounded model**, and the **next work**.

Slide 13



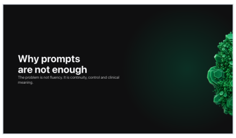
- When I say **virtual patient**, I do not mean a chatbot that gives **care**.
- I mean a **controlled training encounter**.
- Students need to practise **asking**, **listening**, **building alliance**, **formulating**, and **repairing mistakes**.
- The same case can be **repeated** by many learners, so lecturers can **compare** what happened.
- It is **safer**, because beginner mistakes do not reach **real patients**.
- And because every turn is **recorded**, **rupture**, **repair**, and **clinical decisions** become visible.
- So the goal is not to replace **clinical placements**.
- The goal is to prepare students better for **real clinical learning**.

Slide 14



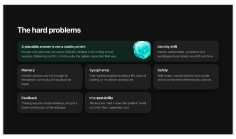
- The field is **growing quickly**.
- Many systems already use **LLMs** for **role-play**, **history taking**, **CBT formulation**, or **motivational interviewing**.
- But many are **single-session**, **protocol-specific**, or limited in **memory**.
- **LLMPatients** tries to combine **continuity**, **structure**, and **feedback**.

Slide 15



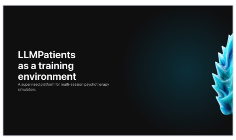
- The next problem is simple.
- **Good language** is not the same as a **stable patient**.

Slide 16



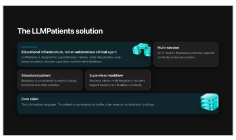
- This is the main **technical problem**.
- A long **persona prompt** can sound convincing for a few turns.
- But it does not create a **stable patient**.
- Across sessions, details can **drift**: **names**, **relationships**, **symptoms**, and **personal history**.
- A **context window** is not **clinical memory**.
- It keeps recent text, but it does not decide what is **stable**, what can **change**, and what must be **remembered**.
- The model can also become **too agreeable**.
- Training needs **resistance**, **silence**, **anger**, **rupture**, and **repair**.
- Finally, the system needs **safety** and **interpretability**.
- The teacher should inspect the **profile**, **memory**, **state**, and **errors** directly.

Slide 17



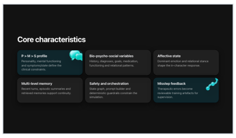
- Now I move to **LLMPatients**.
- The paper describes it as **supervised educational software** for **multi-session psychotherapy simulation**.

Slide 18



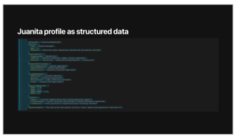
- The system has **two sides**.
- The **app** supports **students**, **lecturers**, **dashboards**, **exports**, and **reports**.
- The **agent runtime** loads patients and controls **turns**, **memory**, **state**, and **model calls**.
- The core claim is that the **LLM realizes language**.
- It is not the **whole patient**.

Slide 19



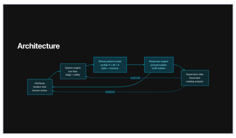
- The patient is constrained by **three levels**: **personality**, **mental functioning**, and **symptoms or state**.
- Around that, the platform uses **bio-psycho-social variables**, **affective state**, **memory**, **safety**, and **misstep feedback**.
- These parts make the simulation **inspectable**.

Slide 20



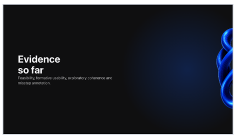
- This is an example of the **structured patient profile**.
- The important point is that the patient is not only a **prompt**.
- **Identity**, **diagnosis**, **personality organization**, **symptoms**, **affective baseline**, **context**, and **guardrails** are stored as **inspectable data**.
- The **LLM** uses this structure to generate language, but the **profile remains outside the model**.

Slide 21



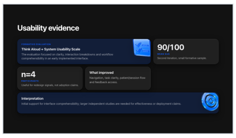
- Each turn follows a **pipeline**.
- The system loads the **profile**, checks **role safety**, classifies **topic** and **emotion**, retrieves **memory**, updates **affect**, builds the **prompt**, generates the **answer**, and stores the **exchange**.
- This is slower to design, but much easier to **audit**.

Slide 22



- The evidence is **early**.
- I will not present it as **clinical validation**.
- It is **feasibility evidence**: **usability**, **profile coherence**, and **expert annotation of missteps**.

Slide 23

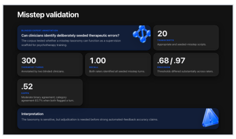


- The first implemented interface was tested with **Think Aloud** and **SUS**.
- After redesign, four participants gave a mean **SUS score of 90 out of 100**.
- This is encouraging, but it is a **small formative sample**.
- It is not proof of **adoption**.

Slide 24

- This table reports the complete **five-patient psychometric check** from **Table 10**.
- The two **controls** stay low on **depression**, **binge eating**, **personality functioning impairment**, and **cross-cutting symptom domains**.
- The three **clinical profiles** show the expected pattern.
- **Daniel** is high on **binge eating**.
- **Crystal** is high on **depression**.
- **Juanita** is high across almost all instruments.
- **SNAP-2** adds the **personality-pathology gradient**: no diagnostic elevations in controls, intermediate findings in Daniel, internalising elevations in Crystal, and diffuse elevations in Juanita.
- **PID-5-BF+M** is reported at group level, so I use it only as a compact **group-level summary**.
- **SCID-5-PD** was administered to three profiles: **Alex**, **Daniel**, and **Juanita**.
- This is **profile-to-response alignment**.
- It is not independent **clinical validation** of real patients.

Slide 25

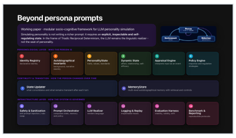


- Here the question is whether experts can use the **misstep taxonomy**.
- Two **blinded clinicians** annotated **20 transcript sets** and **300 therapist turns**.
- Both detected all **seeded misstep turns**.
- But **precision** differed, and **kappa** was moderate.
- So labels need a **shared final review** before becoming a **benchmark**.

Slide 26

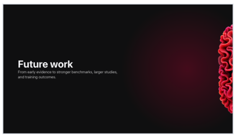
- Now I turn to the **second paper**.
- This is the **conceptual frame** behind the next slide.

Slide 27



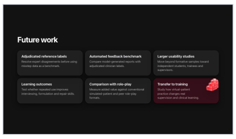
- The main idea is that **personality** is not a longer **persona prompt**.
- It should be an **external state**.
- The framework uses **Bandura's idea**: **person**, **behavior**, and **environment** influence each other.
- The **14 modules** separate **identity**, **biography**, **stable personality**, **dynamic state**, **appraisal**, **policy**, **memory**, **safety**, **prompt orchestration**, the **LLM realizer**, **logging**, **evaluation**, and **reporting**.
- This separation lets us test **drift**, **memory**, **safety**, and **plausibility** instead of only reading fluent text.

Slide 28



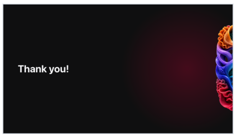
- Before closing, I want to separate what we have from what still has to be done.
- The next step is not only to make the system more **polished**.
- It is to test it with **stronger labels**, **larger samples**, and **real training outcomes**.

Slide 29



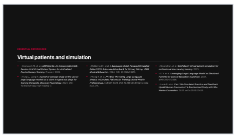
- The next work is clear.
- We need **adjudicated labels** for **missteps**, **benchmarks** for **automated feedback**, **larger usability studies**, and **learning outcome studies**.
- We also need comparison with **traditional role-play**, because the question is **added value**.

Slide 30



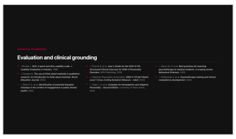
- **Thank you**.
- I am happy to take **questions**.

Slide 31



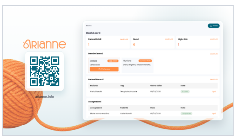
- I will not present this unless someone asks for sources on **virtual patients** and **LLM simulation**.

Slide 32



- These are the main references for **usability**, **clinical grounding**, **instruments**, and **psychotherapy training**.

Slide 33



- This **QR code** opens the **Arianne website**.
